Scaling analysis of the phase-field sublimation model, and how Knudsen transport and surface diffusion would change it

lunar_regolith_DSM `docs/scaling_analysis/scaling_analysis.tex`

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1 The model

Unknowns: ice phase field ϕ (air fraction $1-\phi$), temperature T, vapor density ρ_v. As implemented (`src/assembly.c`, `Residual_A1`),

$$\tau_{\rm sub}\,\frac{\partial\phi}{\partial t} = \varepsilon^2\nabla^2\phi - \phi(1-\phi)(1-2\phi) + \lambda_{\rm sub}\,\phi^2(1-\phi)^2\,\frac{\rho_v-\rho_{vs}(T)}{\rho_{\rm ice}}, \tag{1}$$

$$\rho c_p\,\frac{\partial T}{\partial t} = \nabla\cdot(K\nabla T) + \rho_{\rm ice}L_{\rm sub}\,\frac{\partial\phi}{\partial t}, \tag{2}$$

$$\frac{1}{\xi_v}\,\frac{\partial}{\partial t}\big[(1-\phi)\rho_v\big] = \nabla\cdot\big[(1-\phi)\,D_v\nabla\rho_v\big] - \rho_{\rm ice}\,\frac{\partial\phi}{\partial t}. \tag{3}$$

Equation (1) is the solver's $\partial_t\phi = -3M[\varepsilon^{-1}\phi(1-\phi)(1-2\phi) - \varepsilon\nabla^2\phi] + \alpha_{\rm sub}\phi^2(1-\phi)^2(\rho_v-\rho_{vs})/\rho_{\rm ice}$ with $M = \varepsilon/(3\tau_{\rm sub})$ and $\alpha_{\rm sub} = \lambda_{\rm sub}/\tau_{\rm sub}$. On a regolith wall, $\partial\phi/\partial n = \cos\theta\,\phi(1-\phi)/\varepsilon$. The parameters are set from physical ones:

$$\begin{aligned} \lambda_{\rm sub} &= \frac{a_1\varepsilon\,\rho_{\rm ice}}{d_0\,\rho_{vs}}, \qquad \tau_{\rm sub} = \frac{\varepsilon^2\beta}{d_0}, \qquad d_0 = \frac{\gamma m}{\rho_{\rm ice}k_BT}, \\ \beta &= \frac{1+\delta}{\alpha_c}\,\frac{\rho_{\rm ice}}{\rho_{vs}}\sqrt{\frac{2\pi m}{k_BT}}, \qquad \delta = a_1a_2\,\varepsilon\,\alpha_c\Big(\frac{1}{D_T}+\frac{1}{D_v}\Big)\sqrt{\frac{k_BT}{2\pi m}}, \end{aligned} \tag{4}$$

$a_1 = 5$, $a_2 = 0.1581$. At present D_v is the diffusivity of vapor in air at 1 atm and ρ_{vs} the saturation density of vapor in air.

2 Nondimensionalization

Let R be the length that characterizes the geometry (grain radius or pore width) and T_m the mean temperature; coefficients below are evaluated at T_m. Choose

$$\begin{aligned} \mathbf{x} &= R\,\mathbf{x}', \qquad t = \tau\,t', \quad \tau \equiv \frac{R^2\beta}{d_0}, \\ \rho_v &= \rho_{vs}(T_m)\Big(1+\frac{d_0}{R}\,u\Big), \qquad T = T_m + \Theta\,\vartheta, \quad \Theta \equiv \frac{\rho_{\rm ice}L_{\rm sub}\,d_0}{k_{\rm ice}\,\beta}. \end{aligned} \tag{5}$$

d_0/R is the equilibrium supersaturation of a surface of radius R; τ is the time for such a surface to move a distance R at the velocity $V = d_0/(R\beta)$ the Gibbs–Thomson law gives it; Θ is the temperature rise that conducts away the latent heat released at that velocity ($\Theta = \rho_{\rm ice}L_{\rm sub}VR/k_{\rm ice}$).

Phase field. Write $\rho_{vs}(T) = \rho_{vs}(T_m)(1 + s\Theta\vartheta)$ with $s = \mathrm{d}\ln\rho_{vs}/\mathrm{d}T$. Then $(\rho_v-\rho_{vs})/\rho_{\rm ice} = (\rho_{vs}/\rho_{\rm ice})(d_0/R)\,(u - G_T\vartheta)$ with $G_T \equiv s\Theta R/d_0$, and $\lambda_{\rm sub}(\rho_{vs}/\rho_{\rm ice})(d_0/R) = a_1\varepsilon/R$. With $\tau_{\rm sub}/\tau =$

$(\varepsilon/R)^2$ and $\hat{\varepsilon} \equiv \varepsilon/R$, Eq. (1) becomes

$$\hat{\varepsilon}^2 \frac{\partial \phi}{\partial t'} = \hat{\varepsilon}^2 \nabla'^2 \phi - \phi(1-\phi)(1-2\phi) + a_1 \hat{\varepsilon}\, \phi^2 (1-\phi)^2 \left(u - G_T\, \vartheta\right). \tag{6}$$

Vapor. Expanding the left side of Eq. (3), substituting, and dividing by $D_v \rho_{vs} d_0 / R^3$:

$$\frac{P}{\xi_v}(1-\phi)\frac{\partial u}{\partial t'} = \nabla' \cdot \left[(1-\phi)\nabla' u\right] - \frac{1-m_v}{\Lambda}\frac{\partial \phi}{\partial t'}, \tag{7}$$

$$\Lambda \equiv \frac{L^*}{R}, \quad L^* \equiv \frac{D_v \beta \rho_{vs}}{\rho_{\text{ice}}} = \frac{(1+\delta) D_v}{\alpha_c}\sqrt{\frac{2\pi m}{k_B T}}, \qquad P \equiv \frac{R^2}{D_v \tau} = \frac{d_0}{D_v \beta}, \qquad m_v \equiv \frac{\rho_{vs}}{\xi_v \rho_{\text{ice}}}.$$

(The source coefficient is $\rho_{\text{ice}} R^3/(\tau D_v \rho_{vs} d_0) = \rho_{\text{ice}} R/(\beta D_v \rho_{vs}) = 1/\Lambda$; the m_v term is the $-\rho_v \partial_t \phi$ part of the storage term.)

Heat. Dividing Eq. (2) by $k_{\text{ice}}\Theta/R^2$, with $\hat{c} = \rho c_p/(\rho c_p)_{\text{ice}}$, $\hat{K} = K/k_{\text{ice}}$ and $D_T = k_{\text{ice}}/(\rho c_p)_{\text{ice}}$:

$$Pe_T\, \hat{c}\, \frac{\partial \vartheta}{\partial t'} = \nabla' \cdot (\hat{K} \nabla' \vartheta) + \frac{\partial \phi}{\partial t'}, \qquad Pe_T \equiv \frac{R^2}{D_T \tau}. \tag{8}$$

Wall. $\partial\phi/\partial n' = \cos\theta\, \phi(1-\phi)/\hat{\varepsilon}$.

Groups. Equations (6)–(8) contain:

Group	Meaning	T-dependence	Example[a]
$\hat{\varepsilon} = \varepsilon/R$	interface thickness (numerical)	–	0.02
$\Lambda = L^*/R$	attachment against gas-diffusion resistance	$D_v/\sqrt{T}$	2.8
G_T	latent heat shifting the local saturation	ρ_{vs}/T^2	7×10^{-4}
P/ξ_v	vapor storage (quasi-steadiness)	ρ_{vs}	7×10^{-9}
m_v	vapor mass in the surface balance	ρ_{vs}	9×10^{-4}
Pe_T	heat storage	ρ_{vs}	1×10^{-10}
θ; $k_{\text{air}}/k_{\text{ice}}$; $\hat{c}_{\text{air}}$	contact angle; property ratios	–	–

[a] Air at 1 atm, $-20\,^\circ$C, $R = 50$ μm, $\varepsilon = 1$ μm, $\alpha_c = 10^{-3}$

(`enceladus_DSM/studies/keff_sintering/temperature_scaling/check_groups.py`).

What the scaled model says.

- *All of the strong temperature dependence is in τ.* With Eq. (4),

$$\tau = \frac{(1+\delta)\, R^2 \rho_{\text{ice}}^2}{\alpha_c\, \gamma}\, \frac{1}{\rho_{vs}(T)}\sqrt{\frac{2\pi k_B T}{m}},$$

  so $\tau \propto \sqrt{T}/\rho_{vs}(T)$: an activation energy of $L_{\text{sub}} M_w - \frac{1}{2} R_g T \approx 50$ kJ mol^{-1}. ρ_{vs} cancels in L^*, which retains only $D_v/\sqrt{T} \propto T^{1.31}$.

- *The phase-field equation* (6) *has no physical parameter*: only $\hat{\varepsilon}$ and the constant a_1. This is the calibration doing its job. As $\hat{\varepsilon} \to 0$ it enforces $u = \chi' + v'$ on the surface (curvature χ', normal velocity v'), and Eq. (7) gives $v' = \Lambda\, \partial u/\partial n'$ there.

- *With the small groups dropped*, the physics is $\nabla'^2 u = 0$ in the pore with $u - \Lambda\, \partial u/\partial n' = \chi'$ on the ice, in the time t/τ: one parameter, Λ, plus θ and the geometry. Hence the time–temperature scaling $Q(t;T) = \widehat{Q}(t/\tau(T); \Lambda(T))$ for any geometric quantity Q.

3 Change 1: Knudsen (free-molecular) vapor transport

Why. In air at 1 atm the mean free path is about 0.07 μm, far below the pore size, and Fickian diffusion holds. In vacuum the only gas is the vapor itself; at the vapor pressure of ice below 200 K the mean free path is centimetres or more, so molecules cross a pore without colliding (Kn = mean free path / pore size $\gg 1$).

What changes in the model.

1. *Diffusivity.* $D_v(T)$ is replaced by the Knudsen value for a pore of width d_p, $D_K = \frac{1}{3} d_p\, \bar{v}$ with $\bar{v} = \sqrt{8 k_B T/(\pi m)}$ (Pollard & Present, 1948). It depends on the local pore size, not on pressure; a transition form is $1/D = 1/D_v + 1/D_K$.
2. *Saturation.* ρ_{vs} becomes that of pure vapor, $\rho_{vs} = P_{vs}(T)\, m/(k_B T)$, in place of the mixing-ratio form in air.
3. *Heat.* No gas conduction: $k_{\mathrm{air}} \to 0$ in the pore (radiation aside). The $1/D_v$ in δ becomes $1/D_K$.
4. The attachment law is unchanged: Hertz–Knudsen is already kinetic theory.

Effect on the scaling. Only Λ and the temperature law of τ change. Inserting D_K in L^*, the thermal speed cancels:

$$L_K^* = (1+\delta)\,\frac{d_p}{3}\sqrt{\frac{8 k_B T}{\pi m}}\cdot\frac{1}{\alpha_c}\sqrt{\frac{2\pi m}{k_B T}} = (1+\delta)\,\frac{4\,d_p}{3\,\alpha_c}, \qquad \Lambda_K = (1+\delta)\,\frac{4}{3\alpha_c}\,\frac{d_p}{R}. \tag{9}$$

- Λ_K does not depend on temperature. For constant α_c the time–temperature scaling becomes exact (to the order of the small groups), with $\tau \propto T^{3/2}/P_{vs}(T)$, i.e. $E_a = L_{\mathrm{sub}} M_w - \frac{3}{2} R_g T$.
- Λ_K is large unless α_c is near one: $\Lambda_K \approx 1300\, d_p/R$ at $\alpha_c = 10^{-3}$. Inside a pore the vapor is then uniform and transport is irrelevant; only attachment limits the rate.
- Transport returns over long paths. Across a layer of thickness H the diffusive resistance equals the attachment resistance when $H \sim L_K^* = 4 d_p/(3\alpha_c)$: about 7 cm for $d_p = 50$ μm at $\alpha_c = 10^{-3}$, and 70 μm at $\alpha_c = 1$. Loss of ice to space through a regolith layer is therefore the case where Knudsen transport matters.

Difficulties.

- *A diffusion equation is not valid inside a resolved pore.* Knudsen diffusion describes transport over lengths much greater than the pore size. Within one pore, molecules fly from surface to surface in straight lines; the flux at a point depends on what it can see (as in radiative exchange), which is non-local. A pore-resolving phase-field model with D_K inserted is a closure, not the physics. It is harmless when $\Lambda_K \gg 1$, because the field is uniform anyway, and unreliable when $\Lambda_K \lesssim 1$.
- *d_p is not a field the model has.* It must be supplied: a constant, or a local pore width computed from ϕ (a distance function), which changes as the ice moves.
- *α_c now controls everything*, including whether transport matters at all, and it is the least known parameter.
- *Magnitudes.* ρ_{vs} falls by many orders of magnitude at lunar temperatures; the solver should carry u, not ρ_v. With no gas conduction, heat leaves only through ice and grain contacts (and radiation), so G_T should be re-evaluated with the conductivity of the actual path.

4 Change 2: surface diffusion

What it adds. Molecules migrate along the ice surface from high to low chemical potential. In the sharp-interface description this adds to the normal velocity (Mullins, 1957)

$$v_n = \underbrace{\frac{\sigma - d_0\chi}{\beta}}_{\text{vapor route}} + B\,\nabla_s^2\chi, \qquad B = \frac{D_s\,\delta_s\,\gamma\,\Omega}{k_B T}, \tag{10}$$

with ∇_s^2 the surface Laplacian, D_s the surface diffusivity, δ_s the thickness of the mobile layer and Ω the molecular volume. It conserves ice (it moves it, without exchange with the vapor).

What changes in the model. A conserved, fourth-order term is added to Eq. (1):

$$\tau_{\text{sub}}\frac{\partial\phi}{\partial t} = \big[\text{right side of Eq. (1)}\big] + \tau_{\text{sub}}\nabla\cdot\big[M_s\,g(\phi)\,\nabla\mu\big], \qquad \mu = \frac{1}{\varepsilon}\phi(1-\phi)(1-2\phi) - \varepsilon\nabla^2\phi, \tag{11}$$

with a mobility $g(\phi)$ that vanishes in both bulk phases (for instance $\phi^2(1-\phi)^2$) and M_s calibrated so that the sharp-interface limit gives B (Rätz et al., 2006; Gugenberger et al., 2008). Equations (2) and (3) must use only the non-conserved part of $\partial\phi/\partial t$ as their source, since surface diffusion exchanges neither vapor nor latent heat.

Effect on the scaling. Scaling Eq. (10) as before ($\nabla_s^2\chi = \nabla_s'^2\chi'/R^3$, $v_n = Vv'$):

$$v' = (u - \chi') + S\,\nabla_s'^2\chi', \qquad S \equiv \frac{B\,\beta}{d_0\,R^2} = \frac{\tau}{\tau_s}, \quad \tau_s \equiv \frac{R^4}{B}. \tag{12}$$

- A new group, S, the ratio of the two routes' rates. The reduced problem has two parameters, (Λ, S).
- S depends strongly on temperature. $B \propto D_s/T$ carries the activation energy of surface diffusion, E_s, and τ carries about 50 kJ mol^{-1}, so $S \propto \exp[(E_v - E_s)/R_gT]$. With $E_s \approx$ 24 kJ mol^{-1} (the value Choukroun et al., 2020, attribute to surface diffusion), S grows by a factor of about 6 from -5 to $-40\,^\circ$C and keeps growing as it gets colder.
- $S \propto 1/R^2$: the time scales follow $\tau \propto R^2$ and $\tau_s \propto R^4$ (Herring, 1950), so surface diffusion wins for small grains.
- Time–temperature superposition no longer holds in general: $Q(t;T) = \widehat{Q}(t/\tau;\Lambda,S(T))$. It survives in the two limits, $S \ll 1$ (clock τ, about 50 kJ mol^{-1}) and $S \gg 1$ (clock τ_s, E_s), with a crossover where $S(R,T) = 1$. Cold, fine-grained deposits are on the surface-diffusion side.
- Neck growth laws differ ($w \sim t^{1/3}$ for attachment-limited vapor transport, $t^{1/7}$ for surface diffusion), so the exponent also changes across the crossover.

Difficulties.

- *The coefficient.* $D_s\delta_s$ for ice is uncertain by orders of magnitude and its temperature dependence is not settled, especially near melting where a disordered surface layer forms. S inherits that uncertainty, on top of that of α_c in τ.
- *The sharp-interface limit.* Degenerate-mobility Cahn–Hilliard terms do not automatically reduce to pure surface diffusion; the choice of $g(\phi)$ and a correction to μ matter (Gugenberger et al., 2008). Combining the term with the existing non-conserved dynamics also invalidates the present thin-interface calibration (a_1, a_2, δ), which would have to be redone.

- *Numerics.* The term is fourth order. The quadratic C^1 splines already in use can represent it, but it is stiff (time step limits scale as ε^4 for explicit treatment; the implicit system is harder to solve), and $\nabla_s^2\chi$ is sensitive to how well the neck is resolved, which is already the weak point of these runs.
- *Validation.* With two uncertain coefficients, a neck-growth curve alone cannot separate the routes. Their different dependence on temperature, grain size and saturation is what can.

5 Summary

	Present model	Knudsen transport	With surface diffusion
Groups	$\Lambda(T)$	Λ_K, constant in T	Λ, $S(T)$
Clock	$\tau \propto \sqrt{T}/\rho_{vs}$	$\tau \propto T^{3/2}/P_{vs}$	τ and $\tau_s = R^4/B$
Time–temperature scaling	holds; weak residual through D_v	exact for constant α_c	only for $S \ll 1$ or $S \gg 1$
Grain-size scaling	R^2	R^2	R^2 to R^4
Main difficulty	α_c	non-local transport in resolved pores; d_p	$D_s\delta_s$; calibration; stiffness

References (DOIs checked against Crossref, 2026-10-13). Mullins, W. W. (1957), *J. Appl. Phys.* 28, 333, doi:10.1063/1.1722742. Herring, C. (1950), *J. Appl. Phys.* 21, 301, doi:10.1063/1.1699658. Pollard, W. G. & Present, R. D. (1948), *Phys. Rev.* 73, 762, doi:10.1103/PhysRev.73.762. Rätz, A., Ribalta, A. & Voigt, A. (2006), *J. Comput. Phys.* 214, 187, doi:10.1016/j.jcp.2005.09.013. Gugenberger, C., Spatschek, R. & Kassner, K. (2008), *Phys. Rev. E* 78, 016703, doi:10.1103/PhysRevE.78.016703. Kaempfer & Plapp (2009), Moure & Fu (2024) and Choukroun et al. (2020) as in the project bibliography.